

KENTECH Career Lab

AI-Assisted Development Report

Introduction to AI Programming · KENTECH · Spring 2026

Assignment 4 Submission

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GitHub: <https://github.com/ogidaa0522/kentech-career-lab>

Deployment: <https://kentech-career-lab.vercel.app>

This report describes how AI tools were used throughout the development of KENTECH Career Lab.

Vercel Deployment URL

<https://kentech-career-lab.vercel.app>

GitHub Repository URL

<https://github.com/ogidaa0522/kentech-career-lab>

1. AI Tools Used

Tool	Role
Claude (Anthropic)	Primary coding assistant AND in-product AI engine for question generation and career recommendation
GitHub Copilot	In-editor autocomplete for repetitive code patterns

2. Tasks AI Helped With

- HTML page structure and semantic layout for all four pages
- CSS design system: color variables, dark mode, navbar, card components, loading spinner
- Claude API integration: fetch call, JSON prompt engineering, response parsing
- Interview question rendering — dynamic card creation from API JSON response
- Career recommendation rendering — card layout with rank badge, tags, and next-step highlight
- Dark mode toggle with localStorage persistence

3. Representative Prompts

Prompt 1 — Navbar & Dark Mode

Prompt: "Build a responsive navbar with dark mode toggle for a career guidance site targeting energy engineering students. Use a deep navy background with an orange accent color."

AI Output: Claude generated the full navbar HTML, CSS (including mobile hamburger menu), and JS dark mode toggle with localStorage.

Prompt 2 — Claude API Interview Generator

Prompt: "Write a vanilla JS async function that calls the Claude API with a system prompt and user prompt, parses the JSON array response, and renders each question as a card with question number, type badge, and answer hint."

AI Output: Claude produced the callClaude() utility, showLoading()/hideLoading() helpers, and the full generateQuestions() and renderQuestions() functions.

Prompt 3 — Career Recommendation JSON Schema

Prompt: "Create a Claude API prompt that takes a student's energy interest, grade, work style preference, and strength, and returns a JSON array of 3 career recommendations — each with title, description, personal reason, next step, and tags."

AI Output: Claude designed the system prompt and user prompt template, and produced the recommendCareers() and renderCareers() functions.

Prompt 4 — CSS Pulse Animation

Prompt: "Add an animated energy pulse ring effect to the hero section — concentric rings that expand and fade out, representing energy waves."

AI Output: Claude produced the .pulse-ring CSS keyframe animation and the hero-pulse HTML structure.

4. What I Modified and Improved

• Energy-specific dropdown content

The AI generated generic job fields. I replaced them with 8 KENTECH-relevant energy fields (■■■■■■■, ■■■■, ■■, ■■■■■■■■, etc.) and added 13 real Korean companies, research institutes, and graduate schools organized into optgroups.

• Orange accent color for energy theme

The AI defaulted to a blue-only palette. I added #F97316 (orange) as the accent color to symbolize energy and differentiate Career Lab visually from Study Hub.

• Korean UI copy

All button labels, placeholder text, section headings, and form labels were rewritten in Korean to match KENTECH's student voice and make the tool feel native.

• Structured JSON prompt engineering

The AI's initial career prompt returned unstructured text. I iteratively refined the system prompt to enforce a strict JSON schema with 6 fields per career object, making the output reliably parseable.

- **Copy-to-clipboard feature**

The AI did not include a copy button. I added the copyBtn event listener that formats all questions as plain text and uses navigator.clipboard.writeText().

5. Bugs, Errors, and Limitations Found

- **JSON parsing failure from markdown fences**

The Claude API sometimes returned responses wrapped in ````json ... ```` markdown fences, causing JSON.parse() to throw. Fixed by adding a regex strip: `raw.replace(/^(```json\s*/i, "").replace(/```\s*/i, "").trim()` in the callClaude() utility.

- **Loading spinner not hiding on API error**

When the API call threw an error, clearInterval() was not being called, leaving the loading text cycling indefinitely. Fixed by moving hideLoading(interval) into both the try and catch blocks.

- **Dark mode flash on page load**

If dark mode was saved in localStorage, the page briefly rendered in light mode before JS applied the dark class, causing a visible flash. Fixed by moving the localStorage check to the very top of app.js, before any DOM rendering.

- **Select field not resetting after regenerate**

After clicking 'Regenerate', the form fields cleared unexpectedly on some browsers. Fixed by reading input values at the start of the generate function rather than storing them in module-level variables.